

Experiment analysis step 6: secondary outcomes and HTE

Exploratory secondary outcomes and heterogeneous treatment effects

W241 Plank Study

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0.1 Purpose

This document treats secondary outcomes and subgroup effects as exploratory.

- Secondary outcomes: `RPE`, `motivation`, and `energy`
- HTE targets: `age`, `gender`, and baseline `activity level` as a proxy for fitness

These results are reported after the main crossover estimate and interpreted cautiously.

0.2 Load analytic datasets

```
paired <- sanitize_dt_ascii(fread(derived_path("paired_analysis.csv")))
long_dt <- sanitize_dt_ascii(fread(derived_path("long_repeated_measures.csv")))
long_primary <- long_dt[in_primary_analysis == 1L]

stopifnot(nrow(paired) == 38L)
stopifnot(uniqueN(long_primary$participant_id) == 38L)
```

0.3 Secondary outcomes: paired contrasts

```
secondary_specs <- list(
  list(label = "RPE", high = "rpe_high_tempo", low = "rpe_low_tempo"),
  list(label = "Motivation", high = "motivation_high_tempo", low = "motivation_low_tempo"),
  list(label = "Energy", high = "energy_high_tempo", low = "energy_low_tempo")
)

secondary_rows <- lapply(secondary_specs, function(spec) {
  hi <- paired[[spec$high]]
  lo <- paired[[spec$low]]
  tt <- t.test(hi, lo, paired = TRUE)
```

```

data.table(
  outcome = spec$label,
  mean_high = mean(hi, na.rm = TRUE),
  mean_low = mean(lo, na.rm = TRUE),
  mean_diff_high_minus_low = mean(hi - lo, na.rm = TRUE),
  conf_low = tt$conf.int[1],
  conf_high = tt$conf.int[2],
  p_value = tt$p.value
)
})

secondary_paired <- rbindlist(secondary_rows)
secondary_paired[, `:=`(
  mean_high = round(mean_high, 4),
  mean_low = round(mean_low, 4),
  mean_diff_high_minus_low = round(mean_diff_high_minus_low, 4),
  conf_low = round(conf_low, 4),
  conf_high = round(conf_high, 4),
  p_value = round(p_value, 4)
)]

kable(secondary_paired, caption = "Exploratory paired contrasts for secondary outcomes")

```

Table 1: Exploratory paired contrasts for secondary outcomes

outcome	mean_high	mean_low	mean_diff_high_minus_low	conf_low	conf_high	p_value
RPE	7.2632	6.6842	0.5789	-0.0161	1.1740	0.0562
Motivation	5.1842	4.7105	0.4737	0.0298	0.9176	0.0371
Energy	4.6053	4.3158	0.2895	-0.1748	0.7537	0.2144

0.4 Secondary outcomes: fixed-effects versions

```

run_fe_secondary <- function(data, outcome_var, label) {
  tmp <- copy(data)
  tmp[["participant_id"]] <- factor(tmp[["participant_id"]])
  tmp[["period_factor"]] <- factor(tmp[["period"]])

  fit <- lm(
    reformulate(c("high_tempo", "period_factor", "participant_id"), response = outcome_var),
    data = tmp
  )
  vc <- sandwich::vcovCL(fit, cluster = ~ participant_id, type = "HC1")
  ct <- lmtest::coeftest(fit, vcov. = vc)

  data.table(
    outcome = label,
    estimate = unname(ct["high_tempo", "Estimate"]),
    std_error = unname(ct["high_tempo", "Std. Error"]),
    p_value = unname(ct["high_tempo", "Pr(>|t|)"])
  )
}

```

```

secondary_fe <- rbindlist(list(
  run_fe_secondary(long_primary, "post_rpe", "RPE"),
  run_fe_secondary(long_primary, "pre_motivation", "Motivation"),
  run_fe_secondary(long_primary, "pre_energy", "Energy")
))

secondary_fe[, `:=`(
  estimate = round(estimate, 4),
  std_error = round(std_error, 4),
  p_value = round(p_value, 4)
)]

kable(secondary_fe, caption = "Exploratory fixed-effects treatment coefficients for secondary outcomes")

```

Table 2: Exploratory fixed-effects treatment coefficients for secondary outcomes

outcome	estimate	std_error	p_value
RPE	0.5789	0.4237	0.1803
Motivation	0.4737	0.2905	0.1117
Energy	0.2895	0.3259	0.3803

0.5 HTE setup

We estimate treatment interactions on the primary TTE outcome. The coefficient of interest in each model is the **treatment-by-subgroup interaction**, which asks whether the high-tempo effect changes across participants with different characteristics.

Because subgroup results are exploratory, we report Bonferroni-adjusted p-values across the four planned subgroup tests.

```

hte_data <- copy(long_primary)
hte_data[, participant_id := factor(participant_id)]
hte_data[, period_factor := factor(period)]
hte_data[, age_centered := participant_age - mean(participant_age, na.rm = TRUE)]
hte_data[, male_indicator := as.integer(participant_gender == "male")]
hte_data[, activity_level_ord := as.numeric(activity_level_baseline)]
hte_data[, plank_familiarity_ord := fcase(
  plank_frequency_baseline == 1L, 1L,
  plank_frequency_baseline %in% c(2L, 3L), 2L,
  plank_frequency_baseline == 4L, 3L,
  default = NA_integer_
)]

```

The plank-familiarity proxy comes from the question “Do you regularly perform planks?”. To avoid treating two slightly different “occasionally” codes as distinct levels, we collapse the raw 4-level coding into a cleaner 3-step ordinal variable:

- 1 = rarely or never
- 2 = occasionally
- 3 = regularly

0.6 HTE models

```
run_hte <- function(data, subgroup_var, label) {
  fit <- lm(
    reformulate(
      c("high_tempo", subgroup_var, paste0("high_tempo:", subgroup_var), "period_factor", "participant_"),
      response = "outcome_tte"
    ),
    data = data
  )
  vc <- sandwich::vcovCL(fit, cluster = ~ participant_id, type = "HC1")
  ct <- lmtest::coeftest(fit, vcov. = vc)
  term <- paste0("high_tempo:", subgroup_var)

  data.table(
    subgroup = label,
    interaction_term = term,
    estimate = unname(ct[term, "Estimate"]),
    std_error = unname(ct[term, "Std. Error"]),
    p_value = unname(ct[term, "Pr(>|t|)"])
  )
}

hte_results <- rbindlist(list(
  run_hte(hte_data, "age_centered", "Age"),
  run_hte(hte_data, "male_indicator", "Gender (male vs not male)"),
  run_hte(hte_data, "activity_level_ord", "Activity level / fitness proxy"),
  run_hte(hte_data, "plank_familiarity_ord", "Plank familiarity")
))

hte_results[, p_value_bonferroni := p.adjust(p_value, method = "bonferroni")]
hte_results[, `:=`(
  estimate = round(estimate, 4),
  std_error = round(std_error, 4),
  p_value = round(p_value, 4),
  p_value_bonferroni = round(p_value_bonferroni, 4)
)]

hte_display <- copy(hte_results)[
  ,
  .(
    subgroup = c(
      "Age",
      "Gender",
      "Activity level",
      "Plank familiarity"
    ),
    interaction = c(
      "High tempo x age",
      "High tempo x gender",
      "High tempo x activity",
      "High tempo x familiarity"
    ),
    estimate,
```

```

    se = std_error,
    p = p_value,
    bonferroni_p = p_value_bonferroni
  )
]

kable(
  hte_display,
  caption = "Exploratory HTE interaction results with Bonferroni correction",
  col.names = c("Subgroup", "Interaction", "Estimate", "SE", "p", "Bonferroni p")
)

```

Table 3: Exploratory HTE interaction results with Bonferroni correction

Subgroup	Interaction	Estimate	SE	p	Bonferroni p
Age	High tempo x age	0.2723	0.5090	0.5960	1
Gender	High tempo x gender	5.5517	13.5212	0.6839	1
Activity level	High tempo x activity	-0.8812	5.3176	0.8693	1
Plank familiarity	High tempo x familiarity	2.8831	9.7486	0.7692	1

0.7 Verification

```

stopifnot(nrow(secondary_paired) == 3L)
stopifnot(nrow(secondary_fe) == 3L)
stopifnot(nrow(hte_results) == 4L)

message("Secondary outcomes analyzed: ", paste(secondary_paired$outcome, collapse = ", "))
## Secondary outcomes analyzed: RPE, Motivation, Energy
message("HTE tests analyzed: ", paste(hte_results$subgroup, collapse = ", "))
## HTE tests analyzed: Age, Gender (male vs not male), Activity level / fitness proxy, Plank familiarity

```